

# TCS NQT Tier 3 Masterclass: Pattern Recognition & Visual Geometry

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## Introduction from the Examiner

Welcome to Tier 3. This section covers Pattern Recognition, Visual/Spatial Reasoning, and Geometry (Cubes/Dice/Venn). These questions have a "Moderate" weightage, but they are what I call **Clean Marks**. You can solve them in 30 to 60 seconds if you know the structural rules, giving you extra time for the tough Quant sections.

The biggest mistake students make here is trying to look at the "whole picture." Visual questions are designed to overwhelm your eyes. Below, I have laid out the exact tricks, the specific traps we use to confuse you, and 30 high-priority questions to train your pattern recognition.

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## 1 Visual / Spatial Reasoning & Pattern Recognition

### Tricks, Steps & Shortcuts

- **Single Element Tracking (The Golden Rule):** NEVER look at the whole figure. Pick ONE specific element (e.g., just the shaded circle, or just the arrow) and track its movement from Box 1 to Box 2 to Box 3. Find its rule (e.g., moving  $90^\circ$  clockwise). Eliminate options based on this one element. Then pick a second element and repeat.
- **Degree of Rotation:** Elements usually rotate in standard increments:  $45^\circ$ ,  $90^\circ$ ,  $135^\circ$ , or  $180^\circ$ .
- **Addition/Deletion Rule:** Count the number of lines, dots, or sides in each step. If a shape goes from a triangle (3) to a square (4) to a pentagon (5), the next is a hexagon (6).

### TCS Traps to Avoid

- **The Mirror vs. Rotation Trap:** Sometimes we flip an element (mirror image) instead of rotating it. If an asymmetrical element suddenly points a way that rotation cannot explain, it was flipped.
- **The "Reset" Trap:** In a 5-figure series, sometimes Figure 5 is an exact copy of Figure 1, meaning the series resets every 4 steps. Do not overthink if it looks identical!

## Top 10 High Priority Questions

*Note: Since the exam is visual, these questions describe the visual matrices to train your mental element-tracking.*

1. **Pattern Series:** Figure 1 shows a square with an arrow pointing North. Figure 2 shows the arrow pointing East. Figure 3 shows the arrow pointing South.  
**Question:** Which direction will the arrow point in Figure 4?  
**Solution:** West  
**Explanation:** Track the element: The arrow is rotating  $90^\circ$  clockwise in each step. North  $\rightarrow$  East  $\rightarrow$  South  $\rightarrow$  West.
2. **Pattern Series:** Box 1 has 2 dots and a triangle. Box 2 has 4 dots and a square. Box 3 has 6 dots and a pentagon.  
**Question:** What will Box 4 contain?  
**Solution:** 8 dots and a hexagon  
**Explanation:** Apply the Addition rule. Dots increase by 2 (+2). Shape sides increase by 1 (+1).  $6 + 2 = 8$  dots. Pentagon (5 sides) + 1 = Hexagon (6 sides).
3. **Embedded Figure:** A complex drawing contains intersecting lines forming a star, a hidden circle, and a hidden 'Z' shape.  
**Question:** If you must find the embedded figure, and the options are A) a square, B) a 'Z' shape, C) a pentagon, D) a double arrow, which is correct?  
**Solution:** B) 'Z' shape  
**Explanation:** Embedded figure questions require you to find which simple option is hidden inside the complex master figure without changing its orientation.
4. **Figure Matrix:** Row 1 contains a circle, a divided circle, and a shaded divided circle. Row 2 contains a square, a divided square, and a shaded divided square. Row 3 contains a triangle, a divided triangle...  
**Question:** What is the missing third figure in Row 3?  
**Solution:** A shaded divided triangle  
**Explanation:** The horizontal rule is: Shape  $\rightarrow$  Add dividing line  $\rightarrow$  Shade one half. Apply this to the triangle.
5. **Odd One Out:** Option A is an equilateral triangle. Option B is a square. Option C is a regular pentagon. Option D is a rectangle.  
**Question:** Which figure is the odd one out?  
**Solution:** D) Rectangle  
**Explanation:** Options A, B, and C are "regular" polygons (all sides and angles are equal). A rectangle has unequal adjacent sides, breaking the pattern.
6. **Pattern Series:** A line segment is horizontal. In the next frame, it rotates  $45^\circ$  clockwise. In the next, it rotates  $90^\circ$  clockwise. In the next,  $135^\circ$  clockwise.  
**Question:** What is the rotation for the next frame?  
**Solution:**  $180^\circ$  clockwise  
**Explanation:** The degree of rotation itself is following an arithmetic series: 45, 90 (+45), 135 (+45). The next rotation must be  $135 + 45 = 180^\circ$ .
7. **Water Image:**  
**Question:** If a physical block shows the letter 'M', what does its water image look like?  
**Solution:** The letter 'W'  
**Explanation:** A water image is a vertical flip (upside down). A mirror image is a horizontal flip. 'M' flipped upside down looks like 'W'. (TCS often uses this to trick you in pattern series).

8. **Pattern Completion:** A large circle is divided into 4 quadrants. Top-left has 1 star. Top-right has 2 stars. Bottom-left has 3 stars.  
**Question:** How many stars should be in the bottom-right quadrant?  
**Solution:** 4 stars  
**Explanation:** Reading clockwise or top-to-bottom, the number of elements increases by 1 in each quadrant.
9. **Element Tracking:** In step 1, a black dot is in the top-left corner of a square. Step 2: Top-right. Step 3: Bottom-right. Step 4: Bottom-left.  
**Question:** Where is the dot in Step 5?  
**Solution:** Top-left  
**Explanation:** The dot is moving one corner clockwise per step. In step 5, it returns to the starting position (the Reset Trap).
10. **Alternating Series:** Fig 1 has a '+' sign. Fig 2 has a 'O' sign. Fig 3 has a '+' sign with a dot. Fig 4 has a 'O' sign with a dot. Fig 5 has a '+' sign with two dots.  
**Question:** What is Fig 6?  
**Solution:** A 'O' sign with two dots.  
**Explanation:** The base shape alternates (+, O, +, O, +, O). The modifier (dots) increases every two steps: 0, 0, 1, 1, 2... so the next must have 2.
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## 2 Cuts, Unfolds & Unboxing a Cube/Dice

### Tricks, Steps & Shortcuts

- **The Alternate Box Rule (Open Dice):** When a 3D dice is unfolded into a 2D cross or T-shape, faces that are in a straight line and separated by exactly ONE box are ALWAYS opposites.
- **The Z-Shape Rule:** For open dice, if you can draw a 'Z' (or backwards 'Z') across the grid, the two ends of the 'Z' are opposite faces.
- **The Clockwise Rotation Trick (2 Positions):** If two positions of a dice are given and ONE number is common, write down the numbers starting from the common number in a clockwise direction for both dice. The numbers that align vertically are opposites.
- **Paper Folding (Mirror Rule):** Every time you unfold a piece of paper, you are creating a perfect Mirror Image across the fold line.

### TCS Traps to Avoid

- **The Orientation Trap:** You might correctly deduce that Face A and Face B can be adjacent. BUT, TCS will rotate Face B sideways in the option. Pay attention to the orientation of symbols (like arrows or numbers) relative to the top face!
- **The "Two Common Faces" Trap:** If two dice positions share TWO numbers, the remaining third numbers are definitively opposite to each other. Do not try the clockwise trick here.

## Top 10 High Priority Questions

1. **Q:** Two positions of a standard dice are shown. In position 1, the top is 3, front is 1, and right is 2. In position 2, the top is 3, front is 5, and right is 6. What is opposite to 1?

**Solution:** 5

**Explanation:** Apply the Clockwise Trick. The common face is 3.

Dice 1 clockwise from 3:  $3 \rightarrow 2 \rightarrow 1$ .

Dice 2 clockwise from 3:  $3 \rightarrow 6 \rightarrow 5$ .

Align them: 2 is opposite 6, and 1 is opposite 5.

2. **Q:** An open dice forms a straight vertical line of 4 squares numbered 1, 2, 3, 4 from top to bottom. Attached to square 2 on the left is square 5. Attached to square 3 on the right is square 6. What is opposite to 2?

**Solution:** 4

**Explanation:** Apply the Alternate Box Rule. In the vertical line (1, 2, 3, 4), 1 and 3 are separated by one box, so they are opposite. 2 and 4 are separated by one box, so they are opposite. (5 and 6 are the remaining pair, so they are opposite).

3. **Q:** Two positions of a dice are given. Pos 1: Front=A, Top=B, Right=C. Pos 2: Front=A, Top=C, Right=D. Which letter is opposite to C?

**Solution:** Cannot be definitively found, but A and C are adjacent.

**Explanation:** Trap! Look closely: A and C are visible in BOTH positions. When two faces are common (A and C), the third faces are opposite. Therefore, B is opposite to D. We don't have enough info to find C's opposite, only B and D.

4. **Q:** A square piece of paper is folded in half horizontally, then folded in half vertically to form a small square. A hole is punched in the center of this small square. How many holes will appear when fully unfolded?

**Solution:** 4

**Explanation:** Unfolding vertically creates a mirror image (1 hole becomes 2). Unfolding horizontally mirrors those 2 (2 becomes 4).

5. **Q:** A large cube is painted red on all sides. It is then cut into 64 smaller, identical cubes. How many small cubes have exactly 3 faces painted red?

**Solution:** 8

**Explanation:** Standard formula. Cubes with exactly 3 faces painted are ALWAYS the corner pieces of the large cube. A cube always has exactly 8 corners.

6. **Q:** From the same 64-piece cube above, how many small cubes have NO faces painted?

**Solution:** 8

**Explanation:** Total cubes = 64, which is  $4 \times 4 \times 4$ . This means  $n = 4$ . The formula for 0 painted faces (the inner core) is  $(n - 2)^3$ .  $(4 - 2)^3 = 2^3 = 8$ .

7. **Q:** An open dice is shaped like a 'T'. The horizontal top bar has faces A, B, C. The vertical stem goes down from B with faces D, E, F. What is opposite to B?

**Solution:** E

**Explanation:** Horizontal alternate rule: A is opposite C. Vertical alternate rule: B is opposite E. The remaining faces (D and F) are opposite.

8. **Q:** A circular paper is folded in half (semi-circle), then in half again (quarter circle). A triangle is cut out pointing towards the curved edge. When unfolded, how many triangles are there, and where do they point?

**Solution:** 4 triangles, all pointing outward away from the center.

**Explanation:** Since the triangle points to the curved edge (the outside of the circle), mirroring it during unfolding will place 4 triangles along the perimeter, all pointing outward.

9. **Q:** Three positions of a dice are given. 1) Top: 1, Front: 2, Right: 3. 2) Top: 3, Front: 1, Right: 5. 3) Top: 4, Front: 2, Right: 3. What is opposite to 3?  
**Solution:** 6 (assuming standard 1-6 numbering)  
**Explanation:** Look at who is adjacent to 3. From dice 1: 1 and 2 are adjacent to 3. From dice 2: 1 and 5 are adjacent. From dice 3: 4 and 2 are adjacent. So, 3 is adjacent to 1, 2, 4, and 5. The only number left to be opposite is 6.
10. **Q:** In an open dice, faces 1, 2, 3, 4 form a horizontal line. Face 5 is above 2. Face 6 is below 3. Can faces 5, 2, and 6 be visible together when folded?  
**Solution:** No.  
**Explanation:** Apply the Z-shape rule or logic. 5 is attached to 2, 6 is attached to 3. If you fold this up, 5 and 6 become the top and bottom lids of the box. Top and bottom are opposites. Opposite faces can NEVER be seen together from a single viewing angle.
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### 3 Venn Diagrams

#### Tricks, Steps & Shortcuts

- **Inside-Out Reading:** In geometric Venn diagrams (where a triangle overlaps a circle and a rectangle), identify what each shape represents FIRST. Then, read the intersection. "Inside Triangle AND Inside Circle" means it has the properties of both.
- **The "All vs. Some" Rule:** If circle A is completely inside circle B, it means "All A are B". If they intersect, it means "Some A are B". If they don't touch, "No A is B".
- **Work Backwards:** For word-to-diagram matching (e.g., "Dogs, Animals, Cars"), classify them logically before looking at options. Dogs are a subset of Animals (circle in a circle). Cars have nothing to do with them (separate circle). Find the matching image.

#### TCS Traps to Avoid

- **The "Only" Trap:** If the question asks for "Doctors who are Males", you look for the intersection of Doctor and Male circles. BUT, if it asks for "Doctors who are ONLY Males", you must find the intersection of Doctor and Male, and then SUBTRACT any area that overlaps with a third circle (like 'Adults').
- **The "Not" Trap:** "Find Males who are NOT Doctors." Find the whole Male circle, and completely cross out/ignore any part of it that touches the Doctor circle.

#### Top 10 High Priority Questions

1. **Q:** Which Venn diagram represents the relationship between: *Vegetables, Potatoes, Cabbages*?  
**Solution:** One large circle containing two smaller, non-overlapping circles.  
**Explanation:** Both Potatoes and Cabbages belong to the class of Vegetables (so they are inside the big circle). However, a Potato is not a Cabbage, and vice versa, so their small circles do not touch each other.
2. **Q:** In a diagram, the Triangle represents Teachers, the Circle represents Swimmers, and the Rectangle represents Women. Which region represents "Women Teachers who are NOT Swimmers"?

**Solution:** The area where the Triangle and Rectangle overlap, EXCLUDING any part of the Circle.

**Explanation:** "Women Teachers" = intersection of Rectangle and Triangle. "NOT Swimmers" = must be entirely outside the Circle.

3. **Q:** Which diagram represents: *Musicians, Men, Doctors*?

**Solution:** Three circles overlapping each other in the center (like the Olympic rings or a bio-hazard symbol intersection).

**Explanation:** Some men are doctors. Some men are musicians. Some doctors are musicians. And some people are all three (Men, Doctors, AND Musicians). Therefore, all three circles must partially intersect.

4. **Q:** In a class of 50 students, 30 play Cricket, 25 play Football, and 15 play both. How many play neither?

**Solution:** 10

**Explanation:** Use the formula:  $\text{Total} = n(A) + n(B) - n(A \cap B) + \text{Neither}$ .

$$50 = 30 + 25 - 15 + \text{Neither}$$

$$50 = 40 + \text{Neither} \Rightarrow \text{Neither} = 10.$$

5. **Q:** Which diagram represents: *Earth, Sea, Sun*?

**Solution:** One large circle with a smaller circle inside it, and a completely separate circle outside.

**Explanation:** The Sea is entirely a part of the Earth (small circle inside big circle). The Sun is a separate entity entirely (separate circle outside).

6. **Q:** In a Venn diagram, Circle 1 has numbers 2, 4, 6. Circle 2 has 4, 6, 8. Circle 3 has 6, 8, 10. The number 6 is in the exact center where all three overlap. The question asks for the sum of elements present in exactly TWO circles.

**Solution:**  $4 + 8 = 12$

**Explanation:** Find the numbers that sit in the overlapping regions of only two circles. 4 is shared by C1 and C2. 8 is shared by C2 and C3. (6 is shared by all three, so exclude it). Sum = 12.

7. **Q:** Which diagram represents: *Week, Day, Year*?

**Solution:** Three concentric circles (a small circle inside a medium circle inside a large circle).

**Explanation:** A Day is entirely contained within a Week. A Week is entirely contained within a Year.

8. **Q:** In a survey of 100 people, 60 like Tea, 50 like Coffee. 20 like neither. How many like ONLY Tea?

**Solution:** 30

**Explanation:** People who like at least one =  $100 - 20 = 80$ .

$$\text{Formula: } 80 = 60(\text{Tea}) + 50(\text{Coffee}) - \text{Both}.$$

$$\text{Both} = 110 - 80 = 30.$$

$$\text{People who like ONLY Tea} = \text{Total Tea} - \text{Both} = 60 - 30 = 30.$$

9. **Q:** Which diagram represents: *Professors, Authors, Children*?

**Solution:** Two intersecting circles, with a third separate circle.

**Explanation:** Some Professors are Authors (and vice versa), so they intersect. Children cannot be Professors, and realistically in this logical context, are not classified as Authors, so Children are a separate circle.

10. **Q:** A square represents English speakers, a circle represents Hindi speakers, and a triangle represents Marathi speakers. You are asked to find the number of people who speak English

and Hindi, but NOT Marathi. What do you look for?

**Solution:** The area common to the square and circle, strictly outside the triangle.

**Explanation:** "English and Hindi" means the intersection of Square and Circle. "NOT Marathi" means you must slice away any part of that intersection that falls inside the Triangle (the "Not" Trap).